



VDL ENERGY SYSTEMS

Software update manual – V1.6.x

Additional steps required to migrate from V1.5.x to V1.6.x



1. Introduction

With the introduction of ePU10 software version 1.6.x, non-compatible changes have been added to the workflow. This creates additional requirements to the update workflow of any system operating on an older release version. The standard update sequence will result in a non-operational machine.

The incompatibility of this update is only on software and settings level. All machines operating 1.5.x software are capable of running 1.6.0 after updating the required subsystems.

Compared to the regular software and hmi update procedure, the 1.6.0 update has these additional steps:

- Updating the 4G router firmware and configuration
- For some systems: changing the software variant
- Verifying some system settings

All update steps assume:

- System out of operation with external supply and/or sufficient UPS SOC (>70%)
- Update laptop physically wired to ePU10 internal network or connected via an RMS VPN tunnel.

2. Router update

Before updating the PLC software, the 4G router needs to be updated as well. This requires 4 steps:

- Update the firmware, simply to keep up with the latest security patches.
- Update the configuration, to align the setup of all systems again.
- Roll-out additional services, for the new plc version
- Reload VPN hub settings, so we can reach the system again.

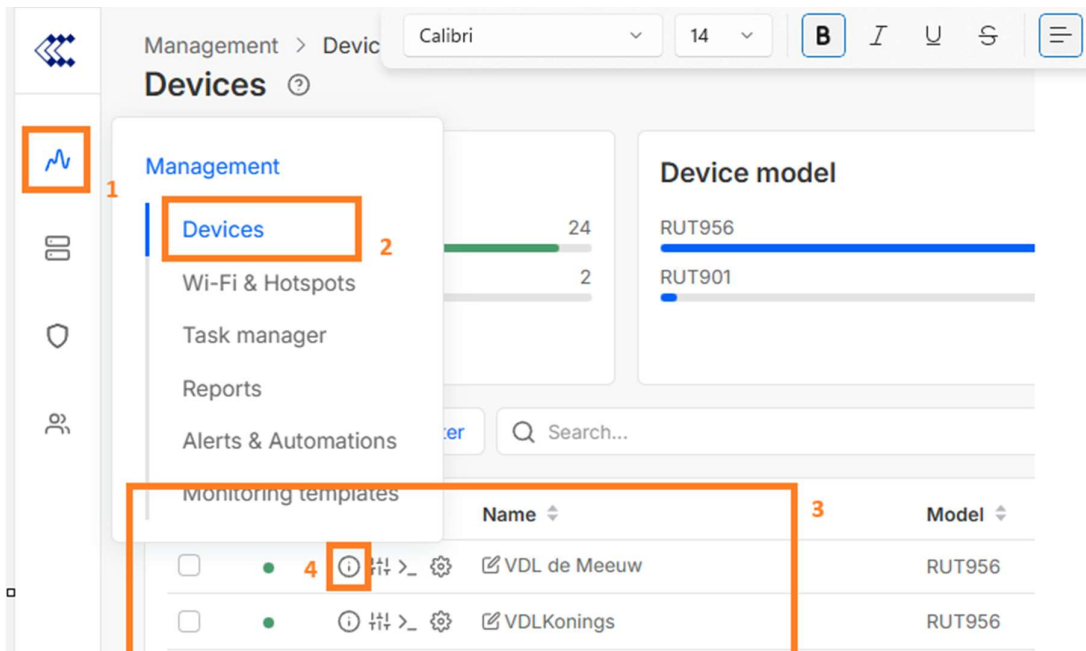
All of these steps can be performed from the web portal of Teltonika:

<https://rms.teltonika-networks.com/>

After login, the individual routers are now listed.

2.1. Firmware update

2.1.1. Go to the device page in the RMS portal



The screenshot shows the Teltonika RMS portal interface. The top bar displays 'Management > Device' and 'Calibri' with a dropdown arrow, followed by '14' and a dropdown arrow, and a toolbar with icons for Bold (B), Italic (I), Underline (U), and a menu icon. The main content area is titled 'Devices' with a help icon. A sidebar on the left contains navigation icons. A dropdown menu is open, showing options: 'Management', 'Devices' (highlighted with an orange box and labeled '2'), 'Wi-Fi & Hotspots', 'Task manager', 'Reports', and 'Alerts & Automations'. Below this, a section titled 'Monitoring templates' is visible, containing a table with columns 'Name' and 'Model'. The table lists two devices: 'VDL de Meeuw' (Model RUT956) and 'VDL Konings' (Model RUT956). The first row of the table is highlighted with an orange box and labeled '3'. The information icon (i) for the first device is highlighted with an orange box and labeled '4'.

2.1.2. On the status page, if a new firmware is available, an icon is shown next to the current firmware version. Click this icon and confirm in the popup window.



VRD2
RUT956
Online

Serial	6001963614	Mobile status
MAC Address	20:97:27:3B:F4:D0	Ethernet ports
IMEI	862430066063187	Connection type
Firmware	RUT9M_R_00.07.15.2	IP Address
Management service	Active	Location
Monitoring templates	2 / 8	

Update to latest stable firmware? X

Latest stable version: RUT9M_R_00.07.19.4

Confirm

Cancel

2.1.3. The firmware will be uploaded and then installed automatically. The connection will be disrupted in this process. Wait for the process to finish and the router to come back online before continuing.

Action status X

No.	Serial	Name	Status
1	6001963614	VRD2	 Uploading firmware... Uploaded 38.94kB of 12.06MB (40.28kB/s) 0.32%

Close

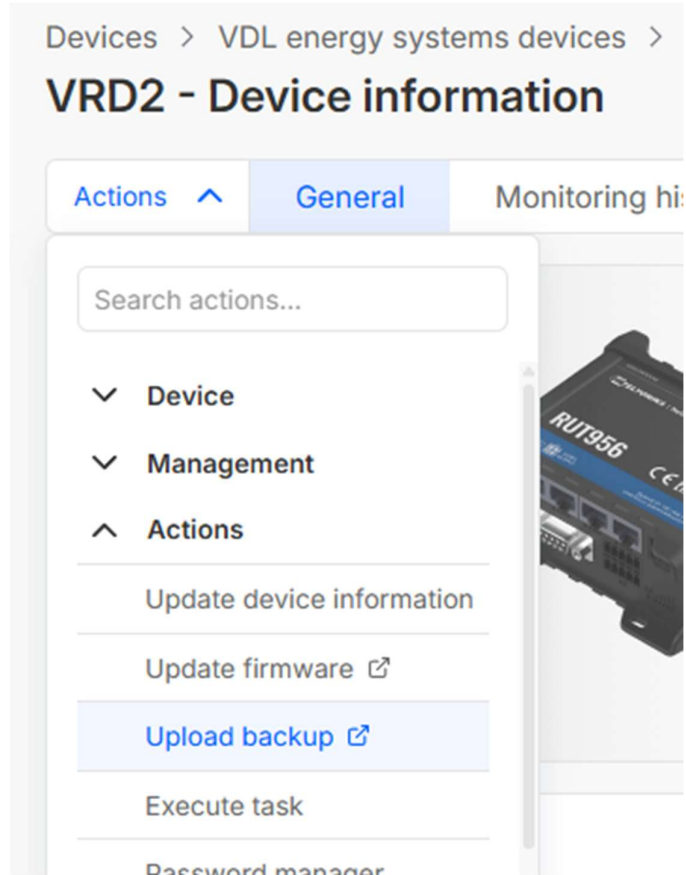
Action status X

No.	Serial	Name	Status
1	6001963614	VRD2	 Successfully started firmware update process.

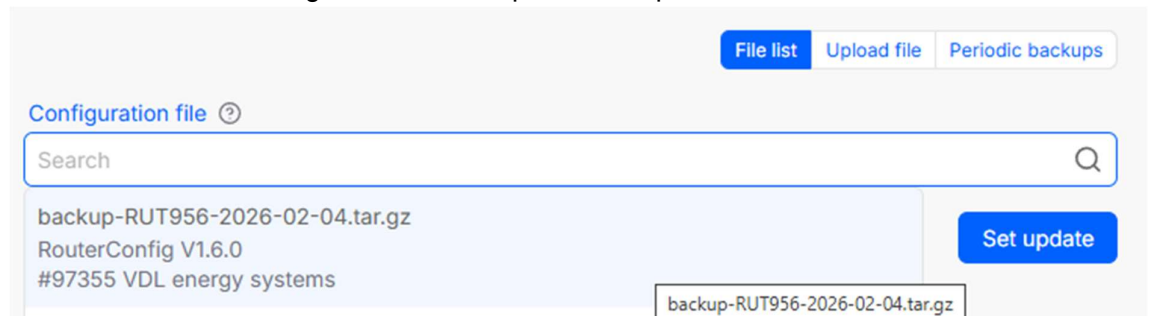
Close

2.2. Configuration update

2.2.1. Similar to the firmware update, the configuration needs to be updated as well. For this, on the status page of the specific device go to actions→Actions→upload backup



2.2.2. Select the Routerconfig V1.6.0 file and press “set update”





2.2.3. Verify if the correct device is selected and press confirm

Backup upload summary

Configuration file can only be applied to devices with identical product code and same or higher firmware version as the file is. Devices with potential conflicts are highlighted in the list below.

Configuration file information

Product code: RUT95620XXXX
Firmware version: RUT9M_R_00.07.19.4
Name: backup-RUT956-2026-02-04.tar.gz
Description: RouterConfig V1.6.0

Verify selection, uncheck device(s) if you wish to exclude any and click on confirm to proceed with the backup upload.

Devices list

☒	VRD2 RUT95620XXXX / RUT9M_R_00.07.19.4	VRD2
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CancelConfirm

2.2.4. Wait for the task to finish

Action status

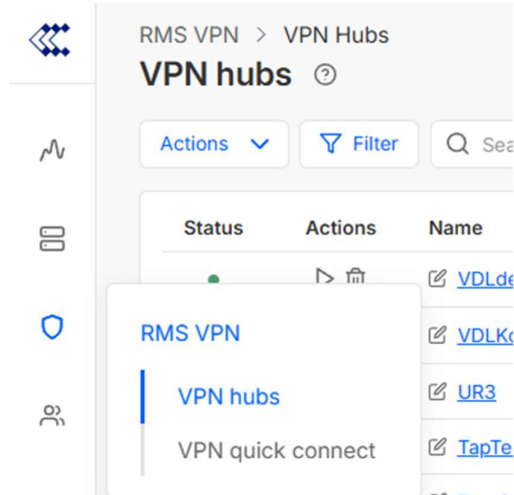
No.	Serial	Name	Status
1	6001963614	VRD2	✓ Successfully uploaded backup.

Close

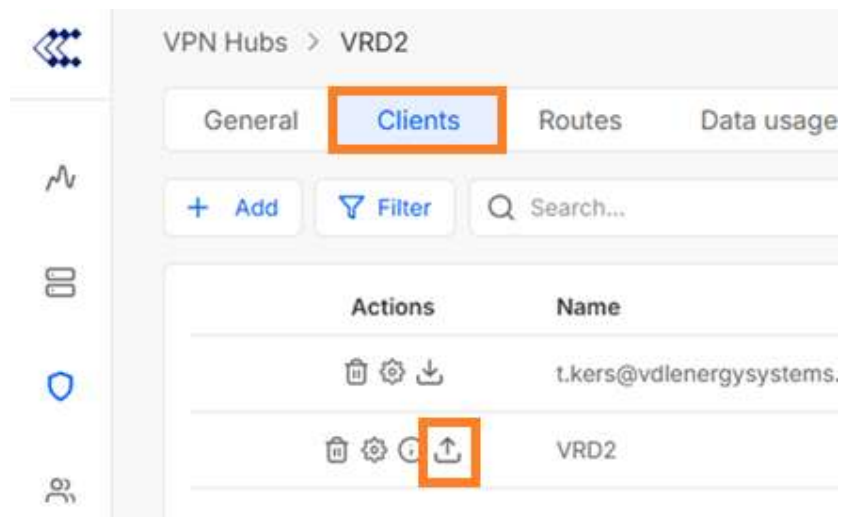
2.2.5. Wait for the device to reboot and come back online. If no customer sim is installed, this can take a bit longer than usual.

2.3. Restart VPN tunnel

2.3.1. The configuration update removes the existing VPN connections to the system, to restart this, go to the VPN hub tab and click the name of the VPN hub that pairs with the machine being updated.



2.3.2. In the hub status overview, go to the Clients tab and press the upload button next to the machine.



2.3.3. Press confirm in the popup window

Resend VPN settings?

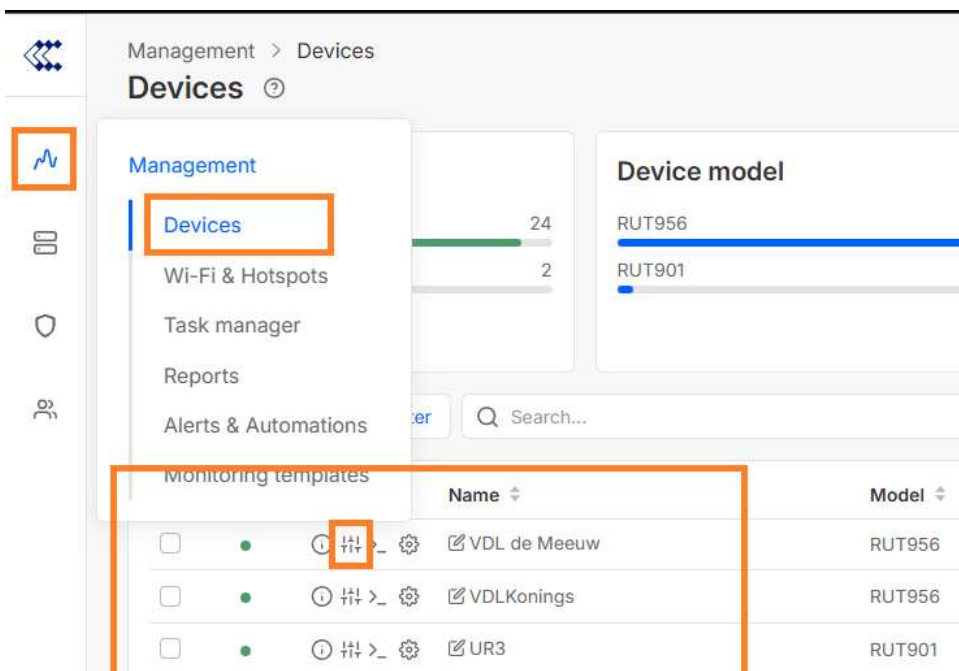


If a device fails to connect to the VPN hub it might be because the device in question does not have the necessary configurations and/or certificates. For example, this could happen after uploading a different backup file, which has no VPN hub settings, or after factory resetting the device. You can attempt to resolve such scenarios by resending the configuration and certificates.

Confirm

Cancel

2.3.4. Go back to the Devices page and press the second button next to the machine to start the WebUI.



Management > Devices

Devices

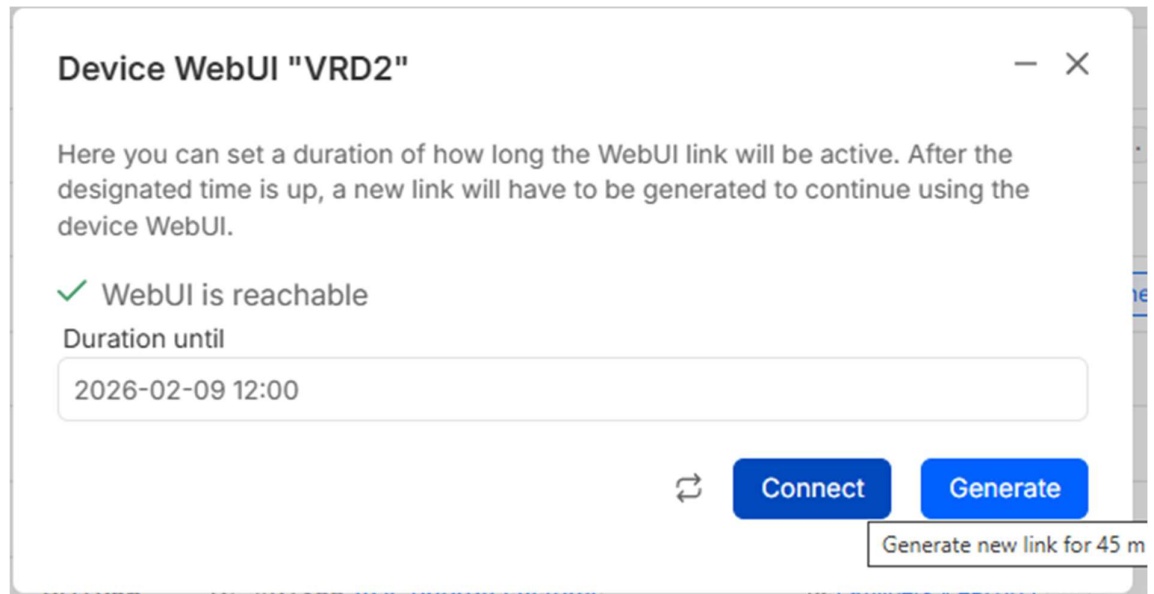
Management

- Devices
- Wi-Fi & Hotspots
- Task manager
- Reports
- Alerts & Automations

Monitoring templates

	Name	Model
☐ 	VDL de Meeuw	RUT956
☐ 	VDLKonings	RUT956
☐ 	UR3	RUT901

2.3.5. Open the connection by pressing “connect” in the popup and wait for the window to open.



2.3.6. Login to router

2.3.7. Go to network→firewall→zones



2.3.8. Edit the zone that has an "RMS_xxxxx" entry



2.3.9. Add Lan and Ext_lan to "allow forward to destination zones" and save settings (twice)

^ "rms_44xY5Y7" zone configuration

General Settings

Advanced Settings

Name * rms_44xY5Y7

Input Accept

Output Accept

Forwarding inside zone Accept

Masquerading ☒ on

MSS clamping ☐ off

Covered networks -- Please select --

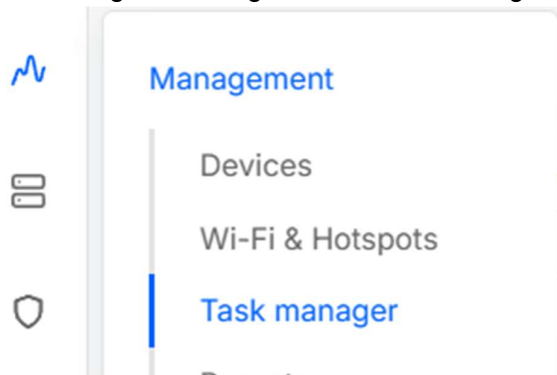
^ Forwarding between zones

Allow forward to destination zones lan x EXT_LAN x

Allow forward from source zones

2.4. Install and enable scripting

2.4.1. In RMS, go to Management → task manager



2.4.2. Press the play button on the "enroll custom scripting" task

☐	Actions	Task group name
☐	   	Enroll custom scripting



2.4.3. Press the play button next to the right machine in the popup window and press “Execute task group” in the popup.

Execute task group

Filter

Search...

Showing 25 of 26 items

Status	Actions	Name
●	▶ ⓘ	VDL de Meeuw
●	▶ ⓘ	VDLKonings
●	▶ ⓘ	UR3
●	▶ ⓘ	VRD2
●	▶ ⓘ	Teunis
●	▶ ⓘ	MCDUtrecht
●	▶ ⓘ	Kuijpers
●	▶ ⓘ	EF9_B8_Preco
●	▶ ⓘ	CWD4
●	▶ ⓘ	CWD3
●	▶ ⓘ	UR2
●	▶ ⓘ	UR1
●	▶ ⓘ	EF8_b_
●	▶ ⓘ	EF7_B6_Weweler

Review task group "Enroll custom scripting"

Before executing a group please, verify that all tasks are as expected and configured properly. Task execution will be carried out in the same order as tasks are provided here. To view full details of any specific task click on the list item below to expand. If a task has any variables created it is also possible to edit them here, but none of the edits will be saved after task group execution, changes are one-time only. To change variables permanently please, use edit mode.

Order no.	Task name	Task type
1	Upload script file	Upload file
2	upload service file	Upload file
3	Set executable	Command
4	Set script to autostart	Command
5	Reboot router	Command

Execute task group

2.4.4. Wait for the task to finish and the router to come back online

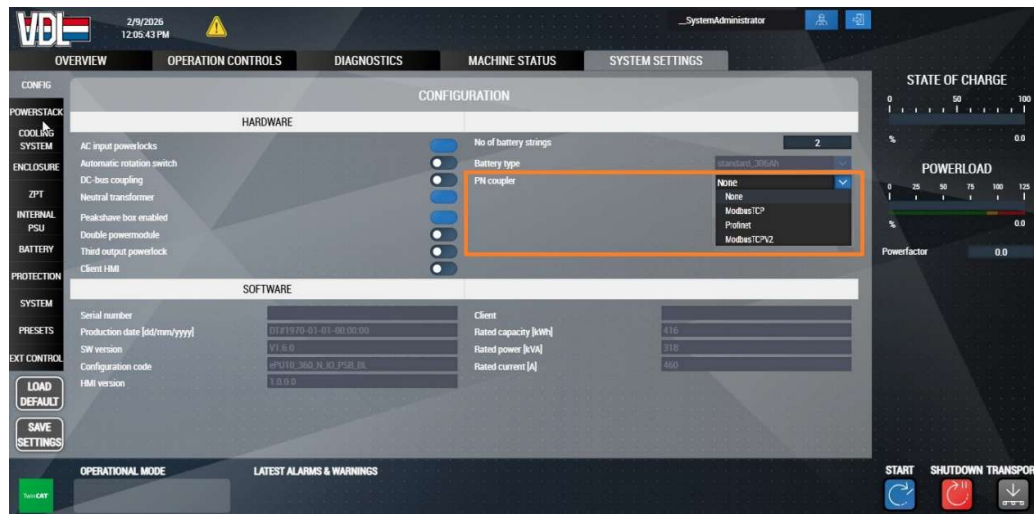
3. PLC update

The Twincat update part is largely similar to previous updates (connect to plc and activate configuration). Two extra things may be required though:

- For machines that use the _MB and _PN hardware variant, they need to be reprogrammed to the _3P_O variant and have some settings changed.
- Machines will assume an LG heatpump being fitted as standard. If a Trane is present, this setting needs to be changed and the plc restarted.

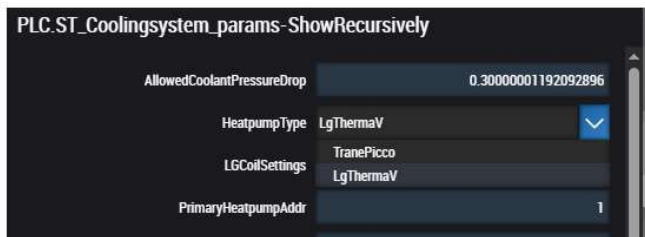
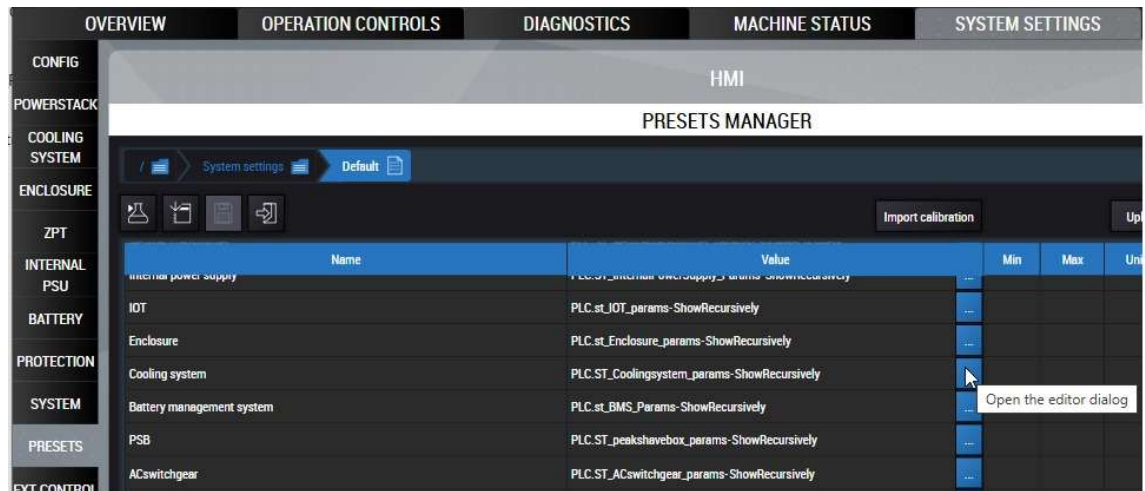
3.1. Change plc variant (Ecofactorij and CWD only)

- 3.1.1. Open V1.6.0 twincat project and connect to the PLC as normal
- 3.1.2. Select the new system variant (3P_O).
- 3.1.3. Verify PLC card configuration (mainly the presence of a led strip card) and activate configuration if correct.
- 3.1.4. After updating, go to system settings/Config in the HMI and select the right external control variant at the “PN coupler” dropdown menu:
 - CWD/AFAS → ModbusTCP
 - Ecofactorij → Profinet
 - United rentals → ModbusTCPV2



3.2. Change heatpump type

- 3.2.1. The heatpump type can be changed in the HMI at System settings/presets/system settings/default/cooling system/HeatpumpType. Select the right version (LGThermaV or TranePicco), save and activate.



3.2.2. If changed, a diagnostics message will be shown that the system needs to be restarted. Either reboot the PLC or powercycle the entire machine to do this.

To reboot the plc, browse to the PLC settings page (10.4.0.40/config) via the RMS tunnel or when directly connected to the 10.4 network. Login, goto Device→ Boot and press reboot.

Wait for the system to start up again and verify the diagnostics message is gone. Also check the cooling system status to verify communication working (only when AC is available, otherwise the heatpump is off).